

# ECEN 662: Estimation & Detection Theory

## Lecture 0: Course Overview

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  - Both  $X$  and  $\theta$  can be *vectors* of *finite* dimensions
- In *statistical inference*, we observe a *realization*  $x$  of  $X$  and need to answer some questions about  $\theta$ , such as:
  - *Estimation*: What is a good *guess* of  $\theta$ ?
  - *Detection*: Is  $\theta \in \Theta_1$ , or is  $\theta \in \Theta_2$ ?

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- A probabilistic model  $f_\theta$  is said to be *identifiable* if

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- Model identifiability is *not* important if our goal is to infer about the likelihood function  $f_\theta$ , i.e., *generative modeling* in machine learning, rather than about the unknown parameter  $\theta$

# Multiple independent measurements

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- A very common situation is where the observed measurements are *independently and identically distributed (iid)*, i.e.,

$$X_i \stackrel{iid}{\sim} f_{\theta}(x_i)$$

so

$$X \sim f_{\theta}(x) = \prod_{i=1}^n f_{\theta}(x_i)$$

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- In practice,  $n$  is always *finite* and inference is always subject to the *randomness* of observation

- Part I: Point Estimation
  - Principles:
    - ▶ Minimum-variance unbiased estimation (MVUE)
    - ▶ Maximum-likelihood estimation (MLE)
    - ▶ Minimum-risk equivariant estimation (MREE)
    - ▶ Bayes estimators
    - ▶ Empirical Bayes estimation
    - ▶ Minimax estimators
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- Part III: Sampling
  - Transform and rejection sampling
  - Monte Carlo integration
  - Markov chain Monte Carlo (MCMC) sampling
  - Diffusion Monte Carlo sampling (if time permits)

# Textbooks, Suggested Papers, and Lecture Slides

- (LC) E. L. Lehmann and George Casella: *Theory of Point Estimation, Second Edition*. Springer 1998
- (LR) E. L. Lehmann and Joseph P. Romano: *Testing Statistical Hypotheses, Third Edition*. Springer 2008
- (RC) Christian P. Robert and George Casella: *Monte Carlo Statistical Methods, Second Edition*. Springer 2004
- *Suggested papers and lecture slides* will be posted on Canvas as we move along

# Course Requirement

- Weekly homework (75%):
  - Theoretical in nature
  - Occasionally *very* challenging
  - Discussions among peers are allowed. However, each student must submit his/her own *independent* work
- Term project (25%):
  - The project can be either a specific application, or some preliminary research on methodology/algorithm
  - The topic of the project needs to be approved by the instructor
- Grading scale: A: 85-100; B: 70-84; C: 60-69; D: 50-59; F: 0-49

# Next lecture

Principle of data reduction and sufficiency